

BRIDGEPULSE · DESIGN DECISIONS REGISTER

BridgePulse design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bridgepulse		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened; open decisions gathered from the decision records, the review note and the build plan work

BridgePulse design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction (foot blocks and clamps, jack block, hub fixing, five penetrations, probe and coupon fixing, cable clips, FieldNode on the square post)	Accept; accept with changes	Accept	Every component of the build plan	BRP-DDR-004, A5
2	Lanyard anchor on the bridge (second retention path in the safety case)	A second, independent beam clamp on the bottom flange; the handrail post; a cross frame where there is one	Independent beam clamp, added to BOM line 4	Lanyard hole in the plate; one bought clamp	BRP-DDR-004, A2
3	Tamper resistance of the FieldNode core on a public footbridge (R9)	Stainless banding with a one-way crimped buckle; worm-drive bands with tamper-resistant screws; a lockable cover	Crimped banding for installations, worm-drive bands on the bench; FieldNode to decide for all post installations	The two band clamps on the post	BRP-DDR-004, A3; BRP-REQ-001, R9
4	Support for the far gauge cable where it crosses between the girders	Along a cross frame or diaphragm; a tensioned catenary wire between two flange clamps; decide at the site survey	Decide at the site survey, preferring a cross member near midspan	Far gauge cable route and the air probe's hanging point; not part of the bench build	BRP-DDR-004, A4
5	FieldNode sensor port pinout and protocol	To be agreed with the FieldNode project; this repo assumes its candidate (pin 1 switched rail, pins 2 and 4 RS-485, pin 3 ground, 5 V)	None yet	Wiring of the M12 panel connector	BRP-DDR-001, O1; BRP-DDR-002, O1
6	First host bridge and co-design partner	Open	None yet	Girder section, plate length, packer length, cable support	BRP-DDR-001, O2
7	Data ownership and what is published beyond monitoring status	Open	None yet	Not part of the build	BRP-DDR-001, O3
8	LoRaWAN region for the first installation	Region of the first host bridge	None in this repo; the 11-byte summary makes R7 independent of it	FieldNode radio and antenna variant	BRP-DDR-002, O4; FND-DDR-001, O1
9			Render aid only		

#	Decision needed	Options	Recommendation	Affects in the build	Source
	Clear window in the hub lid in the appearance model	Render aid only, build with the plain die-cast lid; gasketed window if a status light must be read on site		None (the build uses the plain lid)	REVIEW 2026-09-26, item 1
10	Parts left out of the appearance model (north gauge, air probe, FieldNode and post)	Accept; add them	Accept; FieldNode has its own renders	Renders only	REVIEW 2026-09-26, item 2
11	Cable routes in the appearance model (far cable crossing at +100 mm, FieldNode cable at -130 mm)	Adopt in the model; keep the model's routes	Superseded by BRP-DDR-004: the appearance model should follow the routes in <code>cad/src/model.py</code> when it is next updated	Renders only	REVIEW 2026-09-26, item 3; BRP-DDR-004, P10 and P11

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The host girder's clear web height, flange thickness and root fillet radius	They set the plate length (clear height less 44 mm), the packer length (flange thickness less 0.5 mm) and whether the 22 mm fillet gap and 12 mm chamfer are enough (fillet up to about 20 mm)	BRP-DDR-004, P1 and P2
2	The enclosure's base wall thickness and flatness at the four screw points and eight spacer points	The M5 sealing washers and the countersunk M3 screws need a flat, sound wall of about 4 mm	BRP-DDR-004, P4 and P5
3	Gland clamping ranges against the real cables: M16 for the gauge cables, M12 for the probe leads	Each gland seals only within its range	BRP-DDR-004, P6
4	Flange clip grip ranges: 12.7 mm flange for the cable clips, about 20 mm (flange plus sheath) for the probe clip	A clip outside its range will not hold	BRP-DDR-004, P7 and P11
5	The FieldNode band clamp length (about 300 to 330 mm round the 50 mm square post) and that the FieldNode project accepts mounting without V-blocks	The bands must close with adjustment to spare; the interface is FieldNode's	BRP-DDR-004, P12; BRP-CAL-001, H8
6	The FieldNode core's current cost and mass (\$126.00 and 2.41 kg are copied from FND-CAL-001)	FieldNode's own build plan may have changed them; they feed the complete-monitor cost and the post load	BRP-CAL-001, H1 and K
7	The RP2040 class board's rated temperature range (assumed -20 to +85 °C) and the ADC noise figure (assumed)	They underpin R10 and R4	BRP-CAL-001, D and J

■ Value engineering

Value-engineering target: USD 250.00 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 252.00 (USD 2.00 over the target). The target covers the BridgePulse-specific parts (BRP-DDR-001, D1); the complete monitor with the FieldNode core is USD 378.00.

Main cost drivers (BRP-CAL-001, K): the industrial microSD card (\$20.00), the mounting plate, foot blocks, jaws, packers and jack block (line 4, \$40.00), the strain gauge half-bridges, and the TMP1826 class probes (\$6.00 each). The parts added for construction (BRP-DDR-004) account for \$4.00 of the estimate.

Savings worth trying:

- A consumer high-endurance microSD card rated -25 to +85 °C in place of the industrial card, saving about \$10, if its datasheet covers the range (BRP-DDR-004, A1).
- A cheaper accelerometer breakout that still meets R1 (BRP-DDR-002).
- Quotes at TRL 4, which may move the indicative prices either way.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D11: budget covers the BridgePulse-specific parts, ambient excitation, hourly 10 minute records, hub on the girder and FieldNode on a post, foil half-bridges with dummies, one accelerometer, RP2040 class controller, engineer-only alerts, temperature-compensated trend, shared lab components, pitch unchanged	Amish: "i accept all your recommendations, go with them across all repos."	BRP-DDR-001, BRP-DDR-002
2026-09-25	D12 to D15: load gating with a load regression in TwinKit; 11-byte reduced summary; TMP1826 class probes with an ice-point check; no cost change now, with named savings if quotes come in higher	Amish, same instruction	BRP-DDR-002
2026-09-27	Far gauge cable's vertical runs moved inboard of the bottom flanges in the model	Amish: "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense."	BRP-DDR-003